

แบบฝึกหัด 1 NUMERICAL INTEGRATION AND DIFFERENTIATION I

1. จงหาค่า integration ของ

$$I = \int_2^8 (4x^5 - 3x^4 + x^3 - 6x + 2)dx$$

1.1 โดยใช้วิธี Single Trapezoidal Rule พร้อมคำนวณค่าความผิดพลาด พร้อมเขียนโปรแกรม

1.2 โดยใช้วิธี Composite Trapezoidal Rule $n = 2, 4, 6$ พร้อมคำนวณค่าความผิดพลาด พร้อมเขียนโปรแกรม

2. จงหาค่า integration ของ

$$I = \int_{-1}^2 (x^7 + 2x^3 - 1)dx$$

1.1 โดยใช้วิธี SIMPSON'S RULE พร้อมคำนวณค่าความผิดพลาด พร้อมเขียนโปรแกรม

1.2 โดยใช้วิธี COMPOSITE SIMPSON'S RULE $n = 2, 4, 6$ พร้อมคำนวณค่าความผิดพลาด พร้อมเขียนโปรแกรม

3. จงหาค่า integration ของ

$$I = \int_1^2 \ln x dx$$

3.1 โดยใช้วิธี Composite Trapezoidal Rule $n = 2, 4, 6$ พร้อมคำนวณค่าความผิดพลาด พร้อมเขียนโปรแกรม

3.2 โดยใช้วิธี COMPOSITE SIMPSON'S RULE $n = 2, 4, 6$ พร้อมคำนวณค่าความผิดพลาด พร้อมเขียนโปรแกรม

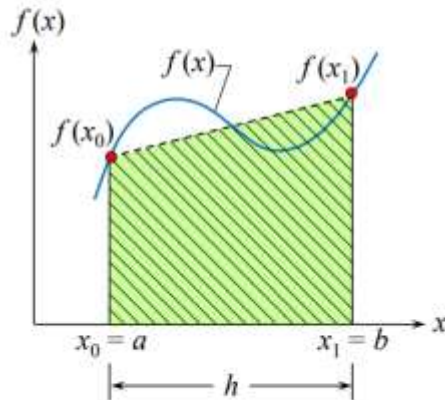
NUMERICAL INTEGRATION AND DIFFERENTIATION

NUMERICAL INTEGRATION TECHNIQUES

- Trapezoidal Rule
- Composite Trapezoidal Rule
- Simpson's Rule
- Composite Simpson's Rule

TRAPEZOIDAL RULE

Integral is approximated by **area under straight line**:



$$I = \int_a^b f(x) dx$$

$$I \approx (x_1 - x_0) \frac{f(x_0) + f(x_1)}{2}$$

$$= \frac{h}{2} [f(x_0) + f(x_1)]$$

Here

$$I = \frac{b-a}{2} [f(x_0) + f(x_1)]$$

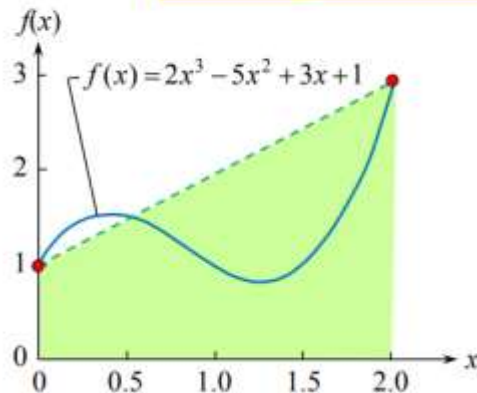
EXAMPLE OF TRAPEZOIDAL RULE

Evaluate $I = \int_0^2 f(x) dx = \int_0^2 (2x^3 - 5x^2 + 3x + 1) dx$

Here $x_0 = a = 0$; $f(x_0) = 0 - 0 + 0 + 1 = 1$

$x_1 = b = 2$; $f(x_1) = 16 - 20 + 6 + 1 = 3$

Thus $I = \frac{h}{2} [f(x_0) + f(x_1)] = \frac{2-0}{2} (1+3) = 4$



Note that the exact integral is:

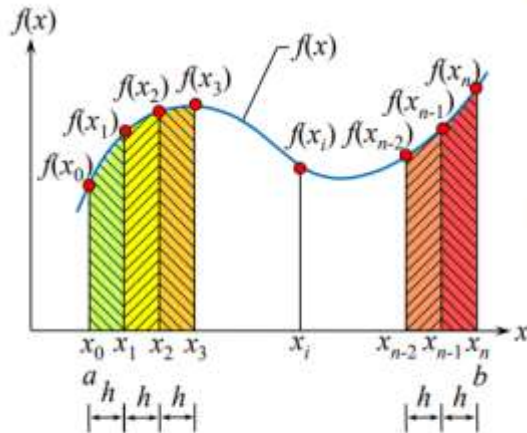
$$\begin{aligned} I &= \int_0^2 (2x^3 - 5x^2 + 3x + 1) dx \\ &= \left[\frac{2x^4}{4} - \frac{5x^3}{3} + \frac{3x^2}{2} + x \right]_0^2 = \frac{8}{3} \end{aligned}$$

Thus the true error is:

$$\varepsilon_t = \frac{(8/3) - 4}{(8/3)} \times 100\% = -50\%$$

COMPOSITE TRAPEZOIDAL RULE

Want to determine $I = \int_a^b f(x)dx$, we divide range a to b into n segments:



Here $h = (b-a)/n$

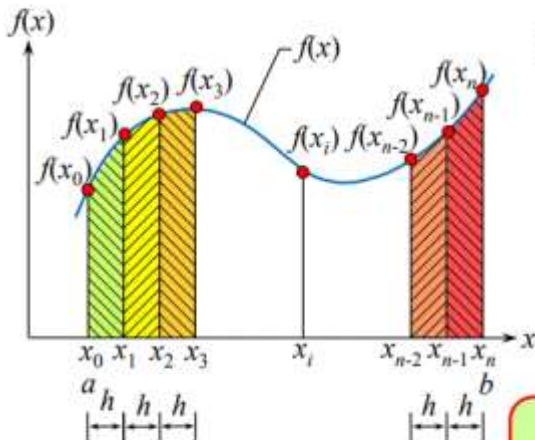
with coordinates for each segment:

$$x_i = x_0 + ih \quad i = 0, 1, 2, \dots, n$$

Thus

$$\begin{aligned} I &= \int_a^b f(x)dx \\ &= \int_{x_0}^{x_1} f(x)dx + \int_{x_1}^{x_2} f(x)dx + \dots \\ &\quad + \int_{x_{n-1}}^{x_n} f(x)dx \end{aligned}$$

$$I = \int_{x_0}^{x_1} f(x)dx + \int_{x_1}^{x_2} f(x)dx + \dots + \int_{x_{n-1}}^{x_n} f(x)dx$$

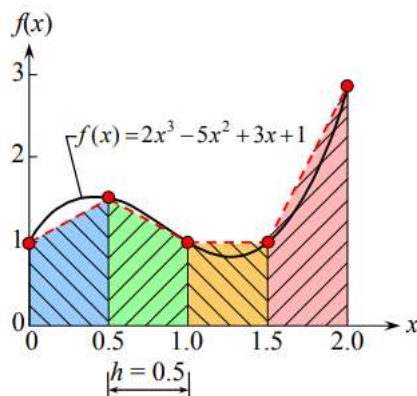


$$\begin{aligned} I &\approx \frac{h}{2} (f(x_0) + f(x_1)) + \frac{h}{2} (f(x_1) + f(x_2)) + \dots \\ &\quad + \frac{h}{2} (f(x_{n-1}) + f(x_n)) \end{aligned}$$

$$\begin{aligned} &= \frac{h}{2} (f(x_0) + 2f(x_1) + 2f(x_2) + \dots \\ &\quad + 2f(x_{n-1}) + f(x_n)) \end{aligned}$$

$$I = \frac{h}{2} \left(f(x_0) + f(x_n) + 2 \sum_{i=1}^{n-1} f(x_i) \right)$$

Example Evaluate $I = \int_0^2 f(x) dx = \int_0^2 (2x^3 - 5x^2 + 3x + 1) dx$ using $n = 4$



Here $h = \frac{b-a}{n} = \frac{2-0}{4} = 0.5$

$$f(x_0 = 0) = 0 - 0 + 0 + 1 = 1$$

$$f(x_1 = 0.5) = 0.25 - 1.25 + 1.5 + 1 = 1.5$$

$$f(x_2 = 1.0) = 2 - 5 + 3 + 1 = 1$$

$$f(x_3 = 1.5) = 6.75 - 11.25 + 4.5 + 1 = 1$$

$$f(x_4 = 2.0) = 16 - 20 + 6 + 1 = 3$$

Thus

$$I = \frac{h}{2} \left(f(x_0) + f(x_4) + 2 \sum_{i=1}^{4-1} f(x_i) \right) = \frac{0.5}{2} (1 + 3 + 2(1.5 + 1 + 1)) = 2.75$$

Exact solution = 2.666667

SIMPSON'S RULE

Integral is approximated by **area under second order Lagrange Polynomial**:

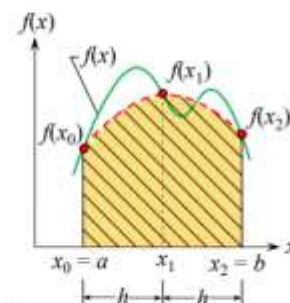
$$I = \int_a^b f(x) dx$$

$$I \approx \int_a^b \left[\frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} f(x_0) + \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} f(x_1) + \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} f(x_2) \right] dx$$

$$I \approx \frac{h}{3} [f(x_0) + 4f(x_1) + f(x_2)]$$

where $h = \frac{b-a}{2}$

or $I \approx \frac{(b-a)}{6} [f(x_0) + 4f(x_1) + f(x_2)]$



EXAMPLE OF SIMPSON'S RULE

Evaluate $I = \int_0^2 f(x) dx = \int_0^2 (2x^3 - 5x^2 + 3x + 1) dx$

Here $h = \frac{b-a}{2} = \frac{2-0}{2} = 1$

$$f(x_0=0) = 0 - 0 + 0 + 1 = 1$$

$$f(x_1=1) = 2 - 5 + 3 + 1 = 1$$

$$f(x_2=2) = 16 - 20 + 6 + 1 = 3$$

Thus $I = \frac{1}{3}[1 + 4(1) + 3] = \frac{8}{3} = 2.666667$

which is exact.

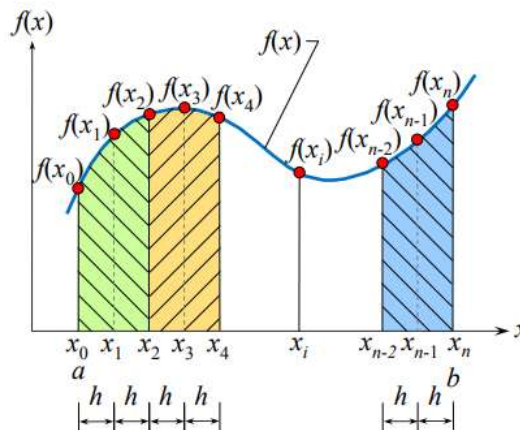
Exact solution = 2.666667

COMPOSITE SIMPSON'S RULE

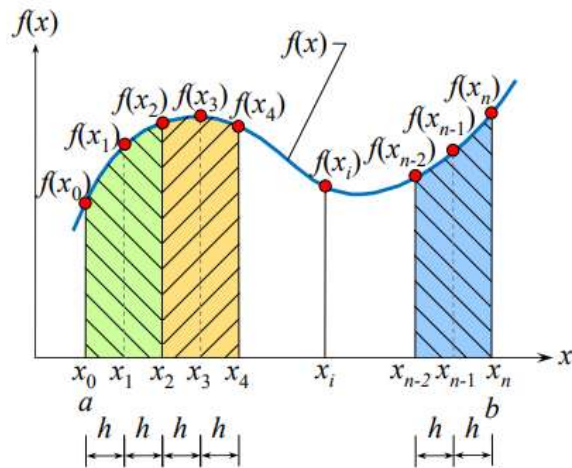
Here $h = \frac{b-a}{n}$; $x_i = x_0 + ih$ $i = 0, 1, 2, \dots, n$

Then

$$I = \int_a^b f(x) dx = \int_{x_0}^{x_2} f(x) dx + \int_{x_2}^{x_4} f(x) dx + \dots + \int_{x_{n-2}}^{x_n} f(x) dx$$



$$\begin{aligned}
 I \approx & \frac{h}{3} [f(x_0) + 4f(x_1) + f(x_2)] + \frac{h}{3} [f(x_2) + 4f(x_3) + f(x_4)] \\
 & + \dots + \frac{h}{3} [f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)]
 \end{aligned}$$



$$I = \frac{h}{3} \left[f(x_0) + f(x_n) + 4 \sum_{i=1,3,5}^{n-1} f(x_i) + 2 \sum_{i=2,4,6}^{n-2} f(x_i) \right]$$

